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Figures/Cover/mpg-logo.pdf

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An atom-photon quantum gate using an atomic clock qubit and a birrefringent cavity

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Abstract

Single atoms coupled to optical cavities provide a powerful platform for photonic quantum information processing. Minimizing the influence of external factors like magnetic field fluctuations is a key goal in this endeavor. In this work, I investigate a novel gate protocol that employs a single ^{87}Rb atom prepared in a magnetic-field-insensitive clock qubit and coupled to a high-finesse birefringent FabryPerot cavity. The cavitys high birefringence creates two well-separated polarization eigenmodes, enabling polarization-selective reflectivity that depends on the atomic qubit state. This allows for the implementation of a controlled phase (CPHASE) gate between the atomic clock states and our photonic qubits. By rotating the basis of the photonic qubit, one is then able to realize an atom-photon controlled-NOT (CNOT) gate.

Show some results here and talk about how well it works and also how one might be able to use the gate

I present simulations based on inputoutput theory and master-equation modeling that quantify the conditional reflection amplitudes, gate truth table, and resulting fidelities under realistic experimental imperfections such as finite mode matching, cavity loss channels, and multiphoton contributions. Our results indicate that atom-state-dependent phase shifts on the photonic qubit are achievable in the current system, providing a viable path toward a robust, high-fidelity, cavity-assisted atomphoton quantum gate.

Contents

Chapter 1

Introduction

Bla bla bla quantum gate important for distant computation between two nodes very cool. Cite some stuff from kimble and duan here [1], talk about recent advancements or stuff that was similar(cite reiserer here [2]).
Basically the idea here is to have a good introduction into the topic. Here I can then talk about:
- what makes up quantum computation - why it's so difficult to scale up in a certain way (maybe look at different examples with large amounts of ^{87}Rb atoms and comment on what the challenges there are when it comes to first of all having lots of them, and them keeping them coherent and stuff like that). - Talk basically a lot about what Duan and Kimble talked about in their introduction of the paper since they were talking a lot about how computing at one node is hard, but spliting it up into smaller nodes and thus making it able to compute stuff. - Then talk about how one is then able to connect the nodes with the gate approach. - Explain what makes it unique (aka the birefringence of the cavity and the ability to then use the clock states which makes it insensitive to external magnetic fields) - Give an outlook over the chapters that are to follow

Chapter 2

Theory

What follows is going to be the introduction and explanation of specific concepts required to understand the key points of this thesis.

2.1 Cavity Quantum Electrodynamics

The field of *Cavity Quantum Electrodynamics* explains the interaction between a resonator (in this case a Fabry-Perot cavity), and a piece of matter with which the light inside the resonator interacts with (here: a ^{87}Rb atom). By placing the atom between two highly-reflective mirrors forming the optical cavity, one increases the amount of times a photon that is inside a cavity bounces back and forth between the two mirrors. Increasing the amount of bounces thus effectively increases the cross-section by the number of round-trips the photon takes, allowing for a strong light-matter interaction.[3]
The theoretical model which explains this interaction in a fully quantum mechanical picture is called the *Jaynes-Cummings Model* [4] and was named after Edwin Jaynes and Fred Cummings who developed it in 1963. For a first look into this model, we'll be considering a three-level-emitter (i.e. our atom), which is interacting with a single cavity optical mode.
The *Jaynes-Cummings* (JC) Model, takes many different quantities into consideration when explaining the interaction and its strength between light and matter.
The photon located in the cavity (which supports only a single optical mode), has a frequency of ω_c. Our atom, which is also located in the cavity, supports a transition frequency close to ω_c, from a coupled state $|c\rangle$ and an excited state $|e\rangle$ at a frequency ω_a, which is described in the dipole approximation. In order to then complete our three-level-emitter picture, we include an uncoupled state $|u\rangle$ with which the photon does not interact due to this state not being coupled to the cavity light mode.
Lastly, any real atom-cavity system, additionally suffers from losses, those mainly being the spontaneous decay of the atomic excitation at a rate of 2γ, and the damping of the light inside our cavity through the absorption or out-coupling by one of the cavity mirrors at a combined rate of κ.
The full Hamiltonian which describes interaction between the atom and the cavity thus reads [5]

$$\mathcal{H}_{\mathcal{JC}} = \underbrace{\hbar\omega_c \hat{a}^\dagger \hat{a}}_{\widehat{=}\text{cavity field}} + \underbrace{\hbar\omega_a \hat{\sigma}^\dagger_{ce}\hat{\sigma}_{ce}}_{\widehat{=}\text{atom}} + \underbrace{\hbar g(\hat{a}\hat{\sigma}^\dagger_{ce} + \hat{a}^\dagger\hat{\sigma}_{ce})}_{\widehat{=}\text{light-atom interaction}} \tag{2.1}$$

The three terms in (2.1) describe the energy of the cavity field, the energy of a bare atom, and the coherent coupling between the two governed by a coupling strength g.
The operators $a^\dagger$ and a are the creation and annihilation operators of the cavity light modes , whereas $\hat{\sigma}_{ce} = |c\rangle\langle e|$ and $\hat{\sigma}^\dagger_{ce} = |e\rangle\langle c|$ are the atomic raising and lowering operators, respectively.
The form of (2.1) assumes the so called *Rotating-Wave Approximation* in which fast oscillating terms at optical frequencies can be neglected.
Solving for the eigenenergies of the system one is left with:

Figures/theory/jc_setup.png

$$E_{\pm}(n) = n\hbar\omega_c + \frac{1}{2}\hbar\omega_a \pm \hbar\sqrt{ng^2 + \frac{1}{4}\Delta_{ac}^2}, \quad n \in \mathbb{N}_0 \tag{2.2}$$

The main take-away from (2.2) is that on resonance ($\omega_a = \omega_c$), the eigenenergies of the system are split by $2\hbar g\sqrt{n}$, giving rise to the so-called *Vacuum Rabi Splitting*.

2.1.1 Birefringent Cavity

Equation (2.1) only covers the case for having a single cavity light-mode. The present example however features a birefringent cavity, which is achieved by physically changing the geometric shape of the cavity mirror during production of the fiber ends.[6] This forces a change in the optical path lengths between different polarizations, leading to a frequency splitting $\Delta\nu$ proportional to the ellipticity ϵ of the mirror surfaces:[7]

$$\Delta\nu = \frac{\nu_{FSR}}{2\pi k}\frac{\epsilon^2}{R_2}. \tag{2.3}$$

The property ν_{FSR} in (2.3) describes the free spectral range, and $k = \frac{2\pi}{\lambda}$ the photon wavenumber. This means that we need to consider a system with 2 different light modes for our JC Hamiltonian, which in our case are going to be the π and the V modes, whereas we're going to be coupled to the π mode, and the V mode will be detuned by ~ 500 MHz. This extra consideration can be included in the Hamiltonian as such:

$$H_{JC} = \hbar\omega_{c,\pi}\hat{a}_\pi^\dagger\hat{a}_\pi + \hbar(\omega_{c,\pi} + 0.5)\hat{a}_V^\dagger\hat{a}_V + \hbar\omega_a\hat{\sigma}_{ce}^\dagger\hat{\sigma}_{ce} + \hbar g(\hat{a}_\pi\hat{\sigma}_{ce}^\dagger + \hat{a}_\pi^\dagger\hat{\sigma}_{ce} + \hat{a}_V\hat{\sigma}_{ce}^\dagger + \hat{a}_V^\dagger\hat{\sigma}_{ce}) \tag{2.4}$$

2.1.2 External Drive

At the heart of the atom-photon gate lies the reflection of the incoming photon off of the cavity system. In order to describe this in a quantum mechanical way, one needs to include a hamiltonian, namely a driving hamiltonian $H_{D,S} = i(\epsilon e^{-i\omega_d t} + \epsilon^* e^{i\omega_d t})(\hat{a} + \hat{a}^\dagger)$. This hamiltonian $H_{D,S}$ is defined in the Schrödinger picture, and has a complex drive amplitude ϵ and driving field frequency ω_d. Adding this extra term to (2.4), and changing into the rotating frame of the driving field, one gets the following Hamiltonian:

$$\begin{aligned} H_{JC} =& \hbar\Delta_{c,\pi}\hat{a}_\pi^\dagger\hat{a}_\pi + \hbar\left(\Delta_{c,\pi} + 0.5\right)\hat{a}_V^\dagger\hat{a}_V + \hbar\Delta_a\hat{\sigma}_{ce}^\dagger\hat{\sigma}_{ce} \\ &+ \hbar g\left(\hat{a}_\pi\hat{\sigma}_{ce}^\dagger + \hat{a}_\pi^\dagger\hat{\sigma}_{ce} + \hat{a}_V\hat{\sigma}_{ce}^\dagger + \hat{a}_V^\dagger\hat{\sigma}_{ce}\right) \\ &+ \mathrm{i}\left(\hat{a}_{\pi,V}\epsilon^* + \hat{a}_{\pi,V}^\dagger\epsilon\right) \end{aligned} \tag{2.5}$$

This then gives a definition of the JC Hamiltonian with respect to an external driving field, and based on the detunings of the driving field with respect to the cavity resonance frequency $\Delta_{c,\pi} = \omega_{c,\pi} - \omega_d$ and the atomic transition frequency $\Delta_a = \omega_a - \omega_d$, respectively.

2.2 Reflection from a Cavity-Atom System

For us it's important that the information we send in via the photon, won't get lost and will get reflected, thus ensuring that our atom-photon gate is suitable for quantum information processing. Therefor when the light enters our cavity, one must ensure by using proper system parameters that the light has maximum transmission through the first cavity mirror (called outcoupling (OC) mirror) relative to the opposing highly reflective (HR) mirror. The highly reflective mirror on the other hand must have maximum reflection so that the information stored in the photon doesn't get lost after subsequent bounces inside the cavity while it's trying to exit via the outcoupling mirror. The total losses can thus be divided into the following parts:

$$\kappa = \kappa_{oc} + \kappa_{HR} + \kappa_{paras} \tag{2.6}$$

The escape probability of our photon is defined by the outcoupling ratio $\frac{\kappa_{oc}}{\kappa}$. This quantity was measured and is $\sim 80\%$.
Another quantity is the optical mode matching between the reflected light at the first cavity and the incoming light (μ_{fr}) aswell as the optical modes of the incoming light and the optical mode inside of the cavity (μ_{fc}).

Our photon that is going to be reflected off of our cavity system can be viewed as an input field. This then can be linked to an output field via input-output formalism. When driving the system only from one side, as it is the case here, the relation between input and output becomes:[8]

$$a_{out}(t) = \mu_{fr} a_{in}(t) + \mu_{fc}\sqrt{\kappa_{oc}}\langle a\rangle(t) \tag{2.7}$$

Here (2.7) was adjusted to also consider the mode matching parameters. Solving Langevin equations for neglidgable atomic excitation and photonic wave envelopes which vary only on a time scale longer than κ, one can write down the analytical representation of the reflection amplitude[9, 10, 11]:

$$r(\omega) = \mu_{fr} - \mu_{fc}^2 \frac{2\kappa_{oc}(i\Delta_a + \gamma)}{(i\Delta_c + \kappa)(i\Delta_a + \gamma) + g^2} \tag{2.8}$$

The reflection amplitude in (2.8) depends on the detuning $\Delta_{c,a} = \omega - \omega_{c,a}$ between the driving laser frequency and the cavity resonance frequency ω_c or atomic transition frequency ω_a.
Include reflection spectrum for different detunings and also phase across different detunings

2.3 Conditional Reflection

Looking at (2.8) one can see that the behavior of the reflection amplitude at zero detuning ($\Delta_a = \Delta_c = 0$) is:

$$r(\Delta = 0) = \mu_{fr} - \mu_{fc}^2 \frac{2\kappa_{oc}\gamma}{\kappa\gamma + g^2} \tag{2.9}$$

In the event of an uncoupled atom-cavity system ($g = 0$), which would be achieved by preparing the atom in the $|u\rangle$ state, the reflection coefficient r is going to be smaller than 0, which would imply a relative phase shift of π between the reflected and incoming light field. Placing the atom in the coupled state $|c\rangle$ however, the reflection coefficient r is going to be bigger than 0 (at sufficiently high coopoerativity). Equipped with this knowledge, that there is a relative phase shift of the output field depending on in which state the atom is, allows the realisation of quantum atom-photon gate with the atom being the controlling qubit and the photon being the target qubit.[1]

2.3.1 Reflection in CPF basis

Changing the nomenclature to match that regularly used in quantum computing, the uncoupled state of the atom is going to be the $|0\rangle$ state, and the coupled one the $|1\rangle$ state. Our photon which as stated before, is going to be either π-polarized or V-polarized, will thus be represented as the input states $|\pi\rangle$ and $|V\rangle$. For the case of our atom being in the $|0\rangle$ state and us sending in a photon which is resonant to the π-mode of the cavity (here: $|\pi\rangle$), the output-field will receive a phase flip, an none for all the other cases either due to being far detuned from the cavity resonance (case of sending $|V\rangle$) and/or atom being in the coupled case $|1\rangle$ and the photon thus getting reflected off of the first cavity mirror:

$$
\begin{aligned}
|0,\pi\rangle &\rightarrow -|0,\pi\rangle \\
|0,V\rangle &\rightarrow |0,V\rangle \\
|1,\pi\rangle &\rightarrow |1,\pi\rangle \\
|1,V\rangle &\rightarrow |1,V\rangle
\end{aligned}
$$

From this we deduce that output from the atom-photon reflection gives us a CPF gate. This conditional phase shift allows the construction of a universal quantum gate that can be transformed into any two-qubit gate using rotations of the individual qubits, which in the case of the photon can be done via retardation waveplates.

2.3.2 Reflection in CNOT basis

Rotating the photonic basis states to

$$
\begin{aligned}
|+\rangle &= \frac{|V\rangle + |\pi\rangle}{\sqrt{2}} \\
|-\rangle &= \frac{|V\rangle - |\pi\rangle}{\sqrt{2}}
\end{aligned}
$$

and keeping the atom either in the coupled ($|1\rangle$) or uncoupled case ($|0\rangle$), results in the following reflection:

$$
\begin{aligned}
|0,+\rangle &\rightarrow |0,-\rangle \\
|0,-\rangle &\rightarrow |0,+\rangle \\
|1,+\rangle &\rightarrow |1,-\rangle \\
|1,+\rangle &\rightarrow |1,-\rangle
\end{aligned}
$$

This then tells us, that simply by using a superposition of our CPF basis, our CPF gate now represents an atom-photon controlled-NOT (CNOT) gate.
Maybe also include brewster plate stuff here, although not quite sure

Chapter 3

Atom-Cavity platform

The system uses a single ^{87}Rb coupled to a birefringent Fabry-Perot cavity inside of an ultra-high vacuum (UHV) chamber. The smaller optical beam path used for the realization of the atom-photon gate, is part of a much larger setup which uses two Fabry-Perot cavities in an X formation, thus giving it the name *CavityX*. For the atom-photon gate, parts of the setup like the magneto optical trap, the 850nm red-detuned trap laser, and the microwave antennas integrated into the UHV chamber, are crucial. Other components, most notably the second cavity, is not within the scope of this thesis, but will be discussed in the outlook.

3.1 Cavity system

At the heart of the CavityX system are the two crossed fiber cavities which are named the "short" cavity (SC) and the "long" cavity (LC), which are distinguishable by their lengths ($79\mu m$ vs. $161\mu m$) as the name suggests, and also by the fact that the short cavity is birefringent. Each cavity fiber is held in place by an in-house built holder shown in figure 3.1.

The incoming light is coupled in and out via the single-mode (SM) mode fiber for either of the cavities. In order to ensure that most of the coupled light exits through the SM fiber, different mirrors coatings are used for the fiber surfaces. The multi-mode (MM) fiber surfaces are coated with a highly reflected dielectric coating, while the opposing SM fiber, has a dielectric coating with a lower reflectivity (10ppm transmission vs. 340 ppm transmission). This leads to most of the coupled-in light to then exit the cavity through which it entered the system.[12]

The cavity fibers are fabricated using a CO_2 laser machining technique by the one of the previous PhD students *Manuel Brekenfeld* in [6].Changing the beam profile of the CO_2 laser allows for the fabiration of either spherical or elliptical surfaces likeit's the case for the short cavity.

Show transmission plot for both the LC and KC (plots are in gianvito's thesis)
The cavity parameters for the short cavity can be taken from the thesis of the current PhD student *Gianvito Chiarella* in [12], except for the mode matching, coupling strength and the linewidth, as these were measuered for a different transition than the one used in this thesis.

3.2 Trapping of a singular atom

3.2.1 Magneto-optical trap

In order to realize an atom-photon quantum gate, one is required to have an atom trapped inside of the cavity with which the photon can then interact with. Therefor one needs to properly position the atom and keep it trapped at the position where it most efficiently couples to the cavity to ensure

Figures/setup/cavities-setup.pdf

Figure 3.1: Experimental setup showing the short and long fiber cavities.[12] look at gianvito's thesis and write down what he did there, also ask him if it's cool to use the figure

Figures/setup/TransmissionProfiles_FiberSurfaces.pdf the most efficient gate operation.

The starting point for this is the trapping of an atomic cloud via a magneto-optical trap (MOT). Once the atomic cloud is trapped, one turns off the MOT, causing the atoms to fall down due to the gravitational force, from which singel atoms can then be loaded into the cavitie's center. The concepts of a MOT are explained in [13].
In order to trap atomic clouds via a MOT, one needs to generate in this case a Rubidium vapor inside of the UHV chamber. This is done via light-induced atom desorption (LIAD), in which atoms are dispensed from a Rubidium dispencer that is placed inside of the vacuum chamber. The light-source is a UV diode *ILH-XQ01-S410* with an emitting wavelength of 410nm, which is pulsed for the length of the time needed to load the MOT.

The MOT loading procedure can be divided into 2 sections:
First comes the loading phase in which the MOT coils, the MOT beams (a cooling laser and a rempumper laser), and the UV light are switched on simaltaneously. This then creates the atomic cloud and cools it down to a temperature close to the ^{87}Rb D_2-line Doppler temperature of 146 μK. [14]
Lastly an optical molasses is applied, which aims at bringing the atom to a temperature much below the Doppler limit. Here the detuning of the MOT beams is increased, whereas the optical power is decreased.
At the end of the MOT cycle, the MOT coils, beams and the UV light are switched off.

Table 3.1: Parameters used for the MOT loading phase.

MOT coil current	23.5 A
UV diode current	140 mA
MOT beams' diameter	12 mm
Loading phase	
Duration	1.4 s
Cooling laser detuning from $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F' = 3$	check detuning again
Repumping laser detuning from $5S_{1/2}, F = 1 \leftrightarrow 5P_{3/2}, F' = 2$	0 MHz
Cooling laser power	20 mW
Repumping laser power	250 μW
Molasses phase	
Duration	1 ms
Cooling laser detuning from $5S_{1/2}, F = 2 \leftrightarrow 5P_{3/2}, F' = 3$	-31 MHz
Repumping laser detuning from $5S_{1/2}, F = 1 \leftrightarrow 5P_{3/2}, F' = 2$	-20 MHz
Cooling laser ramp-down power	1.7 mW
Repumping laser power	250 μW

Parameters used for the MOT loading phase. The UV diode and the MOT coils are permanently on the whole time. The cooling and repumping laser power is measured behind a polarizing beam splitter, which separates the horizontal MOT beam from the top-diagonal ones.

3.2.2 Trapping and positioning of singular atom

At the end of the MOT loading phase, the ultra-cold atomic cloud then sits approximately 10mm above the cavity plane. Turning off the MOT, causes the atoms to fall down towards the cavity due to gravity, until they reach the cavitie's crossing point, where at that time laser beams for cooling and trapping the atom are turned on.
The cooling and repumping laser are shone diagonally onto the crossing point in a counter-propagating configuration. Using orthogonal polarizations in a lin$\perp$lin for the counter-propapagating lasers, the

atoms are cooled to sub-Doppler temperatures via Sisyphus-cooling.
Once the atom is at the center of the cavity, 2 lasers are responsible for keeping it trapped there: The first one is the intracavity laser of the short cavity which has a frequency of 772 nm, and is blue-detuned from the D_2-line. The laser self-interferes in the cavity, forming a standing wave, thus confining the atom at the node i.e. interference minima. Stabilizing the cavity length is achieved by first locking the intracavity laser using the PDH-lock technique and then locking the laser frequency to an optical frequency comb, which serves as a stable frequency reference. For the atom-photon gate the frequency of the laser was chosen so that the cavity is resonant to the D_2-line at 780nm, which is also known.

The second laser is the 852nm vertical trap laser, that's shone in from the top of the cavities' crossing point. The laser is linearly polarized along the long cavity axis, and focussed to a diameter of about $20\mu m$. Afterwards it is then transmitted through the chamber and collimated to a mode diameter of 1cm. Lastly it gets transmitted through a dichroic element, a set of retardation waveplates, a piece of silica glass mounted to a galvanometer scanner (galvo), to then get retroreflected in a cat-eye configuration. Due to the laser being far detuned from the D_2- and D_1-line of the atom, it forms a trapping standing wave, where the atoms are confined in the antinodes- i.e. interference maxima.

In order to achieve maximum cavity coupling, the galvo is scanning the amplitude, thus rotating the silica glass modifies the beam's refraction angle, effectively changing in the optical beam path length, which then moves the antinodes either up or down with respect to the vertical axes of the trap laser. While this is happening, the fluorescence counts coming from the atom during the cooling phase, are collected via the short cavity and detected with superconducting nanowire single-photon detectors (SNSPDs). Live-monitoring the fluorescence counts detected via the SNSPDs which analyzed in real-time with an FPGA that also scans the amplitude of the galvo, one can determine the height and thus the maximum coupling position of the atom by looking at the fluorescence counts. In order to ensure that only a single atom was trapped instead of a two badly coupled one's, the FPGA sequence is as follows:

1. The FPGA starts scanning the voltage of the galvo while monitoring the amount of fluorescence counts detected in a 25ms window coming from the atom.

2. If the fluorescence counts go above a certain threshold (noise-threshold), this means that an atom that's weakly coupled to the cavity is emmitting fluorescence counts into the cavity.

3. The FPGA then continues increasing the voltage, keeping track of the voltage (i.e. position) at which the maximum fluorescence counts were reached, while also adding the amount of clicks in a span of 25ms at each galvo step into a single 32-bit register. This effectively represents an integral over the fluorescence counts vs. galvo amplitude curve.

4. Once the fluorescence counts drop below the (noise-threshold), meaning that the atom has been move out of the cavity center, the integral of the fluorescence counts is evaluated. If the integral is within a certain range, the FPGA will set the voltage at which maximum fluorescence counts were detected. If the integral is out of range (too low: atom was badly coupled anyways; too high: multiple atoms were in one node), the galvo keeps scanning until it either finds a new atom, or reaches the maximum amplitude, causing the MOT and single-atom positioning loop to start again.

A different atom-positioning algorithm was employed in the past (e.g. the one used in *Gianvito Chiarella's* thesis [12]), where the FPGA only scanned up to the point where the maximum fluorescence counts were reached. This past algorithm had the possibility of loading two atoms which were badly coupled to the cavity since there was no way of distinguishing this case from the one of a properly coupled atom. By completing the scan over the entire fluorescence-counts-curve, one is able to properly ensure single atoms are being trapped. Mention something like the g2 or some metric that shows how this algorithm performs and also how it improved over the last algorithm, also do same measurement as gianvito in his thesis to showcase the loading algorithm cuz coool

3.3 Atomic-state preparation

For the atom-photon gate, the single atoms must be prepared in the two ground $m_F = 0$ ground states of the D$_2$-line, those being $5S_{1/2}, F = 1, m_F = 0$ for the uncoupled state, and $5S_{1/2}, F = 2, m_F = 0$ for the coupled state.
The first step in achieving this is to optically pump (OP) the atom into the desired state. This is done repeatedly exciting the atom from a level one wants to depopulate it from to an excited state, from which it will then decay into various ground states, determined by the dipole matrix element of the respective transition. In order to then populate the desired state, one uses a frequency for the pump laser, where all ground states one wants to depopoulate are coupled to the excited state, except for the desired one, gradually populating it over time.
In this work, one first pumps the atom into the $F = 1, m_F = 0$ state. This is done by deploying two laser beams, one which is resonant to the $2 \rightarrow 2'$ transition of the D$_2$-line of ^{87}Rb, which will lead to the depopulation of the $F = 2$ manifold and population of the $F = 1$ manifold. In order to then populate only the $F = 1, m_F = 0$ state, one deploys another beam, resonant to the $1 \rightarrow 1'$ transition of the D$_2$-line. Furthermore the beam must be π-polarized, as in order to only populate the $F = 1, m_F = 0$ state, one needs to uses the fact that the $F = 1, m_F = 0 \rightarrow F' = 1, m_F = 0$ transition is v forbidden.[14]
Include figure showing the actual protocol
Using this technique one is able to populate the $F = 1, m_F = 0$ level with a probability of $\geq 97\%$.[12]

For the atom-photon gate, one also needs to be able to prepare the $5S_{1/2}, F = 2, m_F = 0$ for the coupled state. This is achieved by first preparing the atom in the already prepared $F = 1, m_F = 0$ -state and using a set of microwave antennas integrated into the UHV chamber, to drive the transition via microwave pulses.
Finding the transition frequency is done by performing a microwave spectroscopy measurement, the atom is pumped into the $F = 1, m_F = 0$ via the previously discussed technique. By scanning the frequency of the microwave pulse while keeping the phase fixed, one can measure the population of the F=2 cavity-enhanced fluorescence state detection (SD) for 7.5 μs.
The resulting MW spectrum shown in **??** then shows 3 peaks, which correspond to the $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 0$ and $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = \pm 1$ transitions. From this one can infer that the optical pumping into the $F = 1, m_F = 0$ is done with an efficiency of 97(1)%.[12]
show plot regarding the transition from $F = 1 m_F = 0$ to the other states
Once the frequency for the $F = 1, m_F = 0 \leftrightarrow F = 2, m_F = 0$ transition is known, one can coherently manipulate the atomic population in the ground state. By tuning the MW on resonance, one can drive Rabi oscillations [15] betwee two ground states, where the population evolution in the Zeeman state of $F = 2$ can be modeled as:[16]

$$P_2 = P_1 \cdot \frac{1}{2}(1 - \cos 2\pi\Omega t \cdot e^{-\chi t}) \tag{3.1}$$

where P_1 is the initial population after optical pumping in the state $F = 1, m_F = 0$, Ω is the before determined Rabi frequency, and χ is the damping rate. The result is a combination of a sinusoidal term which describes the coherent state transfer between our two levels, and an exponential factor representing the dissipative mechanism which damps the oscillations.
By scanning the length of the π pulse at a constant MW frequency, one can determine the ideal time it takes for a state to coherently transfer from $F = 1, m_F = 0$ to $F = 2, m_F = 0$. Figure **??** shows the resulting Rabi oscillations for an atom prepared in $F = 1, m_F = 0$ and transfered to the $F = 2, m_F = 0$ state. Fitting equation (3.1), write down fit params

3.4 Laser-beam preparation

For the atom-photon gate, weak coherent pulses are used in order to simulate the effect of single photons getting reflected from the cavity. It's therefore imperative to precisely control the power going into the cavity, and also the frequency which the light possesses. For that specific task the following setup was constructed:

./svg-inkscape/setup_w_detectors_svg-raw.pdf

Figure **??** showcases a double-pass AOM track which allows to offset the frequency of the incoming light by two times the freuqency at which the crystal inside of the AOM is oscillating. The light going into the AOM is blue detuned to the unshifted D_2-line by include actual offset. Part of the laserlight (not shown here), can then be superimposed either with a stable reference frequency source like an optical freuqency comb from the company *MenloSystems GmbH* when the laser frequency doesn't need to be tuned, or with another laser, for when the frequency of the laser needs to be scanned, e.g. for performing a normal-mode spectroscopy to determine the coupling strength g. Regardless of what the other source may be, one can then measure the optical beat signal via conventional photo detectors, and use the beat signal frequency as a feedback for the current of the laser one wants to stabilize the frequency of. The beampath also features two stages connected via polarization maintaining fibers (PM), at which ND filters can be flipped into the path in order to reduce the power giving rise to the weak coherent pulses needed for the gate.

After the AOM track, the light is then coupled into a PM fiber, where it then leads to the so-called *atom table*, at which the atoms are trapped, positioned, and prepared via the aforementioned methods. At the beginning of the track two automatically controlled waveplates allow for the setting of an arbitrary polarization which then gets sent through a set of brewster windows. The reasoning behind he usage of the brewster plates will be explained later. Afterwards a set of compensation waveplates ensures that the H and V polarized light going into the system, are mapped to the π and V mode of the eigenmodes of the cavity. The light then gets reflected off a pellical, which transmits most of the incoming light and reflects a small portion of our light ($\approx 1\%$), onto a mirror which then gets reflected off of a mirror, into the fiber cavity coupler.

The reflected light coming from the cavity then gets reflected off of the previously mentioned mirror, transmitted through the pellical at minimal losses, and coupled into the detection setup. Here, one can set an arbitrary polarization measurement basis via another set of automatically controlled waveplates. It is thus possible to map the reflected light in two modes which then get detected via two SNSPDs.

3.4.1 Brewster window Array

Brewster windows are a piece of uncoated UV fused silica, when placed at the so called brewster angle, that the p-polarized portion of the incomming light gets transmitted without any losses, while the s-polarized portion of the light gets partially reflected.
Using the system parameters and calculating the reflection amplitude via (2.8) it was apparent that the reflection from the atom-cavity system, introduces extra polarization dependent losses in the π mode as opposed to the V mode. Although the effect on the reflection in the CPF basis wouldn't be

Figures/setup/brewster_mount_image.pdf

of any concern, it would lead to a reduced fidelity when measuring in the CNOT basis, as there the two polarization modes would need to have the same amplitude in reflection in order to show the expected flip from one state to the other.
In order to combat this and keep keep the fidelity high, an array of brewster plates were installed on a custom machined mount, which place the brewster plates at the brewster angle relative to the incoming light, so that the V polarized portion of the incoming light gets attenuated prior to interacting with the atom-cavity system. This ensures that in reflection, both polarization modes have the same reflection amplitude, allowing for the observation of the flip on the photonic qubit.
The brewster angle can be simply calculated via:[17]

$$\tan \theta_B = n_t/n_i \tag{3.2}$$

$$n_t^2 - 1 = \frac{0.6961663\lambda^2}{\lambda^2 - 0.0684043^2} + \frac{0.4079426\lambda^2}{\lambda^2 - 0.1162414^2} + \frac{0.8974794\lambda^2}{\lambda^2 - 9.896161^2} \tag{3.3}$$

In (3.3) n_t is the optical density of the brewster window, and n_i would be the optical density from which the incoming light is coming from (here: air $n_i \sim 1.004$)

3.4.1.1 Custom brewster window mount

The custom brewster window mount as mentioned before has the main functionality of placing the elements in an array at the brewster angle θ_b relative to the incoming light. It can house up to 12 brewster windows when using the screw whole in the middle or 8 brewster windows, which allows the mount to be placed on two seperate posts, making the overall mount more stable. The latter is the configuration eventually used in the setup, as alongside a more stable setup, the ideal fidelity laid between 6 and 7 brewster windows, which would make adding more windows useless change that.
Furthermore, the brewster window mount has two sections in which the layout of one half is mirrored to that of the other half. The reason behind this is that at each transmission through the brewster window, the optical beam path gets offset, which significantly reduces the coupling into the cavity fiber. In order to combat this, the other half of the brewster window placement was mirrored so that one half shifts the beam path up (relative to the direction of the into which the brewster window is pointing to come up with a better way to explain this), and the other shifts it back down. Although this is only the case for when the total amount of brewster windows is even and both halves are balanced, at most the shift would be equal to that of a single brewster window shifting the optical beam path, which is able to be compensating by realigning the input bea into the cavity fiber.

3.4.1.2 Characterization of brewster windows

For the characterization of the brewster windows themselves, two measurements were conducted. For the first one, only a singular brewster plate was inserted into the custom mount to check the average attenuation across a set of brewster windows in the V polarization. This ensures that all brewster windows operate up to spec, and that the attenuation of the V polarization is happening in constant jumps. The second one, places multiple brewster windows in an array to closely resembe the setup used in the actual experiment. With this one can then used the measured powers in transmission both for H and V to determine the ideal amounts of brewster windows one would need to use in order to achieve the highest gate fidelity.

Figures/setup/characterization_of_brewster_windows.pdf

3.4.2 Pellicle Beamsplitter

Before the construction of the current beam path setup, the only way to couple light, other than that of the intracavity laser, into the cavity, was done via the so called *probe beam*. The light coming from the probe beam would be coupled into the cavity-fiber, through the transmission of one and subsequent reflection off of another dichroic mirror. Each of these mirrors is a include product names here from the company include company name here, and possess the feature of transmitting most of the incoming light and reflecting only a small portion. This setup was chosen so that both the intracavity light trap and the probe beam could be coupled into the fiber simaltaneously.

The problem with this way of coupling light into the cavity, for a beam where precise control over the polarization and it's losses is crucial, was the fact that in reflection the dichroic mirrors were introducing additional polarization dependent losses in the s and p polarization. An earlier setup chose to still couple light into the cavity when reflecting off of the dichroic mirror, while including a set of compensation waveplates to map the H and V polarization of the incoming light to the s and p polarizations of the dichroic mirror, and then another set of compensation waveplates right in front of the fiber cavity coupler to map s and p to π and V. However with no means to check the polarization of the reflected light from the dichroic mirror via a polarimeter, due to special constraints this proved to be unreliable.

Finally by changing the placement of the beam coupler for the beam, and instead opting to use a pellicle, which had much smaller polarization dependent losses in reflection, one was able to circumvent the issue. Furthermore, one was able to very that the polarization dependent changes both in reflection and transmission through the pellical were minimal, thus ensuring that the correctly reflected light won't receive further penalties when going to the detectors.
include numbers that showcase the losses for each polarization and stuff for the dichroic mirrors and also for the pellical (which are much smaller but still).

Chapter 4

Photonic Qubit Path

4.1 Polarization-Dependent Losses

4.2 Brewster Window Array

4.2.1 Custom Brewster Window Mount

4.2.2 Characterization of Brewster Windows

4.2.3 Simulated Impact on Gate Fidelity

4.3 Coupling into Cavity Fiber via Pellicle Beamsplitter

4.4 Optical Beam Path

4.5 Reflection and Normal-Mode Spectroscopy

Chapter 5

Atom-Photon Gate

5.1 Characterization of system parameters

5.1.1 Cavity coupling and losses

5.1.2 Brewster plate attenuation

5.2 Reflection from cavity

5.3 Atom-Photon gate performance

5.3.1 CPF and CNOT gate truth tables

5.3.2 Average gate fidelity

Chapter 6

Summary and Outlook

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